A Controlled Analysis of Physics-Informed Augmentation for Powder X-Ray Diffraction Classification

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***Abstract*: Models trained on simulated powder X-ray diffraction (pXRD) patterns can perform well on synthetic benchmarks while struggling with measured data. We study whether physically motivated augmentation improves space-group classification while holding a SIMPOD-based Swin-T model and input representation fixed. The corrected benchmark contains 48,913 and 48,892 structures in two training folds and 24,496 synthetic test structures, matched to the original authors' published identifiers. The clean two-fold ensemble reaches 82.92% synthetic top-5 accuracy, numerically close to the published Swin result of 82.69%. Augmentation reduces synthetic top-5 accuracy to 79.63% but increases top-5 accuracy on 20 experimental patterns from 6/20 to 11/20; a class-prior predictor reaches 7/20 on the same patterns. Additional fold-0 runs also favor augmentation, with gains of two and three patterns. Leave-one-component-out comparisons identify broadening and background as the largest contributors on a shared augmented validation set. The primary paired comparison gives an exact two-sided McNemar p-value of 0.0625. Augmentation therefore produces a consistent transfer gain across the evaluated runs for this classifier, while broader experimental generalization requires independent validation.**

***Index Terms*: powder X-ray diffraction, space-group classification, simulation-to-experiment transfer, Swin Transformer.**

I. Introduction

Powder X-ray diffraction is widely used to characterize crystalline materials because diffraction peak positions and intensities carry information about their underlying structure [1]. Automated prediction of crystal symmetry can help narrow the candidates considered during analysis. Two difficulties make space-group classification particularly challenging. Closely related structures can produce almost identical one-dimensional patterns, and practical measurements introduce variation that ideal simulations never contain. Open crystallographic resources such as the Crystallography Open Database (COD) [2] make large simulated training sets possible. However, high accuracy on simulated patterns does not by itself establish reliable performance on experimental scans. Peak broadening, background scattering, intensity distortions, counting noise, and incomplete angular coverage can change the input distribution. A useful evaluation must therefore distinguish numerical reproduction of a synthetic benchmark from transfer to measured data.

This study uses the SIMPOD benchmark of Rincón et al. [3] as a reference and analyzes how a fixed, physically motivated augmentation pipeline affects the transfer of one space-group classifier. We retain a Swin-T architecture and the released radial-image representation, reproduce the available portion of the published split, and compare clean and augmented training under the same primary checkpoint-selection rule. Our contributions are:

1) A controlled comparison of clean and augmented training on the corrected published-identifier split, with architecture, representation, split and selection rule held fixed, reporting synthetic accuracy and paired per-pattern experimental outcomes together.
2) A leave-one-component-out ablation of five augmentation effects, in which independent random streams keep the surviving effects' parameter draws unchanged and every arm is selected on one shared augmented validation set.
3) Robustness analyses of the experimental result: repeated training runs, an alternative checkpoint-selection rule, two conventions for unmeasured angular ranges, a background-subtraction control, stratification by angular coverage, and a class-prior baseline.
4) A reproducibility record: a documented split-allocation pitfall that silently removes common space groups from training, and a verification package from which every reported number can be recomputed.

The resulting evidence favors augmentation on the available measured patterns; Section VI discusses the limits of this finding.

II. Related Work

Early work established that neural networks can infer crystal symmetry from diffraction patterns. Park et al. [4] applied convolutional models to simulated diffraction data, while Vecsei et al. [5] studied neural classification of crystal symmetries. Oviedo et al. [6] demonstrated the value of domain-informed augmentation for small XRD datasets. Tiong et al. [7] combined pattern shaping with a multistream architecture to improve classification from noisy patterns. These studies show that both the representation and the training distribution can materially affect performance.

Subsequent research has examined how synthetic training data should reflect experimental variability. Schuetzke et al.

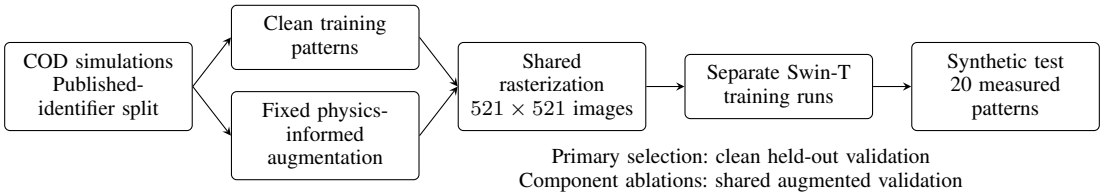


Fig. 1. Clean COD simulations and their fixed augmented counterparts are rasterized to the same 521×521 representation and used to train separate Swin-T models. Only training folds are augmented. Primary checkpoints are selected on clean held-out-fold validation data; component ablations use a shared augmented validation set.

[8] studied training-data design for powder-diffraction phase identification. Lee et al. [9] addressed generalizability and perturbation robustness together, and Salgado et al. [10] investigated deep learning on large diffraction datasets, including adaptation toward experimental conditions. Multiphase identification, as studied by Lee et al. [11], is a related but distinct task from assigning a single space-group label.

SIMPOD [3] provides an open simulated benchmark and evaluates several neural architectures, including a pretrained Swin model. Our clean baseline is a controlled reproduction using the available published identifiers; differences in the COD snapshot and optimizer prevent interpreting it as an exact rerun of the original experiment.

The recent AlphaDiffract preprint [12] predicts crystal system, space group and six lattice parameters with a one-dimensional ConvNeXt, a loss that penalizes errors by distance in the maximal-subgroup graph, and a ten-model ensemble. It is trained on GSAS-II simulations of 312,267 ICSD and Materials Project structures, each rendered 100 times with sampled instrumental and sample broadening, with Poisson and Gaussian noise drawn again for every sample during training. The simulations span 2θ from 5° to 20° at 20 keV, corresponding to d-spacings of 7.1–1.8 Å, whereas the Cu Kα range used here covers 17.7–1.1 Å. On 734 background-corrected RRUFF patterns, the ensemble reaches 66.2% space-group accuracy against 3.27% for a majority-class baseline. Differences in task, angular window, training data, metric and evaluation set make a direct accuracy comparison uninformative: that figure counts exact matches on 734 patterns, while our primary endpoint is top-5 accuracy on 20. The two studies are complementary in method. AlphaDiffract applies augmentation throughout as a design choice, without a clean-trained control or per-component ablation, whereas this study holds architecture, representation and split fixed to measure what augmentation contributes and which components carry the effect. The opXRD database [13] provides an additional route toward broader experimental validation, although it was not used in the present experiments.

III. Methods

Fig. 1 summarizes the data, training and evaluation workflow.

A. Data and Benchmark Split

Clean patterns were simulated from COD crystal structures using Dans_Diffraction 3.4.0 with Cu Kα radiation (wavelength 1.5406 Å), a 2θ range of 5–90°, 10,824 grid points, and a nominal peak width of 0.01°. Intensities were normalized by their maximum. Matching the original authors' split identifiers to the available corpus yielded 48,913 structures in fold 0, 48,892 in fold 1, and 24,496 in the test set, for 122,301 structures in total. Approximately 98% of the intended identifiers were available.

The folds contain no overlapping COD identifiers. A separate structure-key audit nevertheless identified 285 duplicate-structure groups shared between the combined training folds and test set, covering 310 test rows. Consequently, this evaluation measures performance under that allocation and is not a fully structure-disjoint test of generalization. Results from an earlier defective split (Section V) and from earlier runs that selected checkpoints on the test set are excluded from this manuscript.

B. Input Representation and Classifier

Each one-dimensional pattern was downsampled by nearest-neighbor selection to 1,250 values and converted to a 521 $\times$ 521 radial image, using the released rasterization settings (coordinate extent ± 260, $k = 5$, and offset 20). The downsampled length follows the released code rather than the 1,024-point description in the benchmark paper. The grayscale image was repeated across three channels and scaled by 1/255, with no resizing, cropping or ImageNet channel normalization, matching the released pipeline.

The classifier was torchvision's Swin-T [14], initialized with IMAGENET1K_V1 weights. Its final layer was replaced with a linear map from 768 features to 230 outputs. International Tables space-group numbers 1–230 were mapped to class indices 0–229. This architecture and image-processing path were kept fixed across clean training, augmented training, and component ablations.

C. Physics-Informed Augmentation

Augmentation was applied to the one-dimensional intensity pattern before rasterization, in the order broadening, intensity reweighting, background addition, angular shift, and counting noise. Intensities were then normalized again. Only training folds were augmented; synthetic test patterns were never perturbed. One augmented rendering was generated per

structure before each training run and reused at every epoch, seeded deterministically from the run's generator seed and the structure identifier so that each rendering can be regenerated exactly. Thus, augmentation was offline and fixed within a run; additional training seeds used separate rendering draws. Independent random-number streams for the individual effects preserved their parameter draws when one component was disabled.

Broadening was always enabled in the full augmentation configuration. The instrumental full width at half maximum (FWHM), H, followed a Caglioti-type dependence [15]. A constant size-related FWHM term of 0.02–0.30° and a strain term of $4\varepsilon|\tan\theta|(180/\pi)$, with ε in [0, 0.005], were combined in quadrature with the instrumental width. Gaussian filtering approximated the resulting angular dependence using 16 blended segments.

Intensity reweighting was activated with probability 0.4 and used a March–Dollase-shaped function [16] with r in [0.7, 1.0]. The implemented weighting depends on detected-peak index distances, not reflection indices and a physical orientation axis; it is therefore a texture-inspired surrogate. Background was added with probability 0.9 using a degree-2 or degree-3 polynomial reaching 5–30% of the peak maximum. In 30% of those cases, a broad halo was added with center 15–35°, FWHM 10–20°, and amplitude up to 15% of the peak maximum.

A global angular shift was applied with probability 0.7, uniformly within ±0.05°. Poisson counting noise was applied with probability 0.9, with the maximum-intensity count scale sampled log-uniformly between 500 and 5,000.

D. Training and Checkpoint Selection

All runs used AdamW [17] with learning rate 10^{-4}, weight decay 0.05, label smoothing 0.1, and batch size 32. A three-epoch linear warmup preceded cosine decay over a 30-epoch cap. Validation used a fixed 10,000-row subsample of the held-out fold, drawn independently of the training seed. The best checkpoint maximized validation top-5 accuracy, and early stopping used patience five, a minimum improvement of 0.15 percentage points, and no stopping before epoch 12. Synthetic test scores were logged every epoch but never selected a checkpoint or stopped a run reported here.

For the primary clean–augmented comparison, both models selected checkpoints on the same clean validation distribution. Seed 0 included reciprocal fold-0 and fold-1 training, followed by an ensemble that averaged the models' logits before ranking labels. Seeds 1 and 2 repeated fold-0 training only.

The component study trained fold-0 models with each augmentation effect removed in turn. Every arm selected on the same augmented fold-1 validation draw. A clean comparator used the checkpoint with the best score on that augmented validation set, saved during the primary clean run; its training duration was still governed by clean-validation stopping.

The primary seed-0 models and three ablation arms (full augmentation and the removals of broadening and background) were trained on an NVIDIA H100 NVL, the other three ablation arms on an NVIDIA B200, and the seed-1 models and the seed-2 clean model on an NVIDIA RTX 4090, all with bfloat16 mixed precision. The seed-2 augmented model was trained from scratch on Apple MPS in full precision, replacing an interrupted cloud run that was discarded.

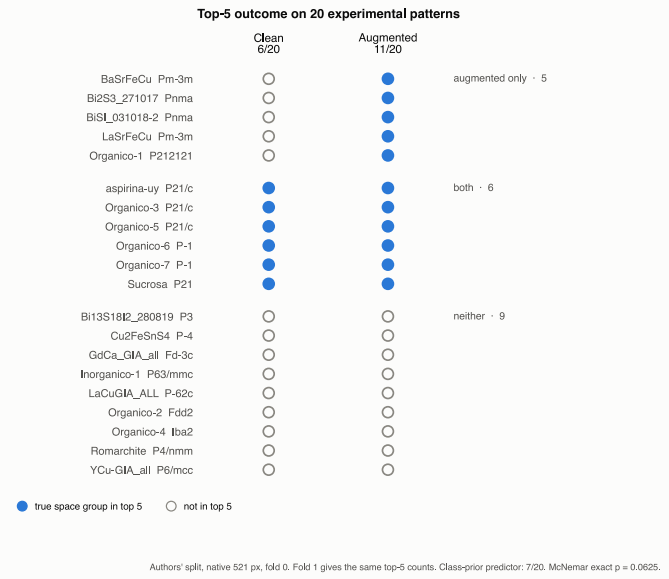


Fig. 2. Paired top-5 outcomes on the 20 experimental patterns for the primary seed-0 fold-0 models. Six patterns are correct for both models, nine for neither, and five only after augmented training; no pattern is correct only for the clean model.

E. Experimental Evaluation and Statistics

The experimental set comprised 20 measured patterns supplied privately by the SIMPOD authors. Labels were converted from Hermann–Mauguin symbols to International Tables numbers with Dans_Diffraction, the package used to simulate the training patterns, so experimental and training labels follow the same convention. Patterns were interpolated onto the simulation grid, max-normalized, and rasterized through the same input path. Primary evaluation extended the nearest measured boundary value outside each scan's angular range. Only three patterns covered the full 5–90° interval; zero filling was evaluated as a sensitivity check.

Top-k accuracy was the fraction of patterns whose true space group appeared among the k highest-scoring labels, with top-5 as the primary endpoint and top-1 reported separately. A class-prior baseline always returned the five most frequent training labels: 14, 2, 15, 19, and 4. We report experimental counts alongside percentages, nominal Wilson 95% intervals [18], and exact two-sided McNemar tests [19] on paired top-5 correctness.

IV. Results

A. Synthetic Benchmark and Experimental Transfer

Table I reports the primary seed-0 results. The clean two-fold ensemble achieved 44.13% synthetic top-1 and 82.92% top-5 accuracy. Its top-5 result is 0.23 percentage points above

TABLE I
PRIMARY PERFORMANCE AND PUBLISHED REFERENCE

Model / reference	Synthetic top-1 (%)	Synthetic top-5 (%)	Experimental top-1	Experimental top-5
Published Swin [3]	44.98	82.69	10%[a]	35%[a]
Clean ensemble	44.13	82.92	2/20 (10%)	6/20 (30%)
Augmented ensemble	39.23	79.63	4/20 (20%)	11/20 (55%)
Class-prior baseline	N/A	N/A	3/20 (15%)	7/20 (35%)

[a]Published experimental scores are provided for reference. Identity of the experimental sets could not be confirmed, so these are not paired comparisons with our results.

TABLE II
COMPONENT ABLATION: SEED 0, FOLD 0

Training variant	Validation top-5 (%)	Δ (pp)	Synthetic top-5 (%)	Experimental top-5
Clean, alt. selection	72.70	−5.23	78.85	7/20
Full augmentation	77.93	0.00	78.98	11/20
Without broadening	74.42	−3.51	77.80	9/20
Without background	75.17	−2.76	78.00	7/20
Without noise	76.23	−1.70	78.54	10/20
Without zero shift	77.92	−0.01	79.03	10/20
Without texture	78.08	+0.15	79.31	9/20

All rows use the same 10,000 augmented validation patterns. Δ is validation top-5 minus the full-augmentation score, in percentage points. The clean row uses its augmented-validation-selected checkpoint. Experimental counts use boundary-value filling; each ablation arm is a single run.

the published Swin reference, while top-1 is 0.85 points lower. This numerical agreement supports the corrected baseline as a useful comparison point, without establishing superiority over the original experiment on a different available corpus.

Full augmentation lowered ensemble synthetic top-5 accuracy by 3.29 percentage points, to 79.63%, while increasing experimental top-5 accuracy from 30% (6/20) to 55% (11/20). Both single-fold augmented models also scored 11/20, compared with 6/20 for each clean model.

For the primary fold-0 paired comparison (Fig. 2), both models were correct on six patterns, both were incorrect on nine, and augmentation added five successes with no clean-only successes. The exact two-sided McNemar p-value was 0.0625. Nominal Wilson intervals were approximately 14.5–51.9% for 6/20 and 34.2–74.2% for 11/20. A class-prior predictor that always returns the five most frequent training labels reaches 7/20 (35%) top-5 on our experimental set, one pattern above the primary clean models, which gives useful context for both results.

B. Contribution of Individual Augmentation Components

Table II and Fig. 3(a) compare the component ablations using a shared augmented validation set. Full augmentation reached 77.93% validation top-5 accuracy. Removing broadening produced the largest decline (3.51 percentage points), followed by background (2.76 points) and counting noise (1.70 points). Removing zero shift changed accuracy by −0.01 points, while removing the texture surrogate increased it by 0.15 points.

The zero-shift and texture removals changed validation top-5 by only −0.01 and +0.15 percentage points, respectively, which is small relative to the observed run-to-run variation (validation top-5 of the three augmented fold-0 replicates spanned 0.84 points on the clean held-out rows), and a model jointly omitting both was not trained.

TABLE III
REPEATED FOLD-0 EXPERIMENTAL EVALUATION

Seed	Clean top-5	Aug. top-5	Δ (pp)	McNemar p
0	6/20	11/20	+25	0.0625
1	7/20	9/20	+10	0.625
2	7/20	10/20	+15	0.375

All pairs are evaluated on the same 20 patterns. Δ is the increase in experimental top-5 percentage points. Seed 2 includes a different device and precision for the augmented run.

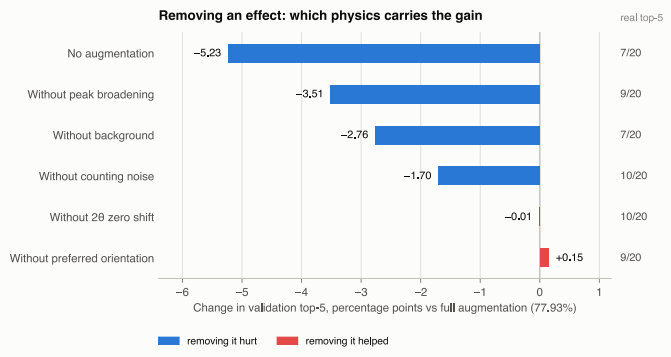
Removing background reduced top-5 performance from 11/20 to 7/20; removing broadening reduced it to 9/20. Removing noise, zero shift, or texture yielded 10/20, 10/20, and 9/20, respectively. These results identify broadening and background as useful components in this setup without proving that the other effects are unnecessary.

Checkpoint selection also changed the clean comparator. Selecting its epoch-2 checkpoint on augmented validation increased experimental top-5 accuracy to 7/20, reducing the clean–augmented gap from five patterns to four.

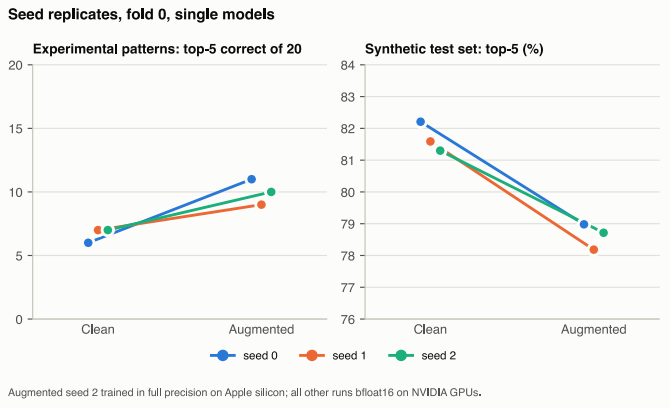
C. Repeated Runs and Preprocessing Sensitivity

Augmentation improved experimental top-5 accuracy in all three fold-0 run pairs (Table III, Fig. 3(b)). Gains were five, two, and three patterns, corresponding to 25, 10, and 15 percentage points. Synthetic top-5 accuracy remained lower for each augmented model: clean scores were 82.21%, 81.59%, and 81.30%, compared with 78.98%, 78.18%, and 78.71% after augmentation.

Changing the unmeasured-range fill from boundary values to zero left both primary augmented single-fold models at



(a) Component ablation



(b) Repeated runs

Fig. 3. Augmentation-component and repeated-run results. (a) Changes in shared augmented-validation top-5 accuracy relative to full augmentation (77.93%), with experimental top-5 counts at right. The no-augmentation row denotes the clean comparator with alternative checkpoint selection; preferred orientation denotes the texture-inspired surrogate. (b) Experimental top-5 counts and synthetic top-5 accuracy for clean and augmented fold-0 models across seeds 0–2. The panels summarize distinct comparisons and use separate axes.

11/20. The clean fold-0 model increased from 6/20 to 7/20, while fold 1 remained at 6/20. A separate SNIP background-subtraction control [20], implemented with 40 iterations in log–log–square-root space, produced clean scores of 6/20 and 7/20 and augmented scores of 9/20 and 11/20. This particular preprocessing control did not recover the augmented models' performance for clean training, but it does not rule out other preprocessing approaches.

Angular coverage did not concentrate the gains in the most extrapolated scans. The augmented fold-0 model was correct within its top five for 1 of 3 fully covered patterns, 3 of 4 at 88–99% coverage, 5 of 11 at 76–82%, and 2 of 2 at 59–65%, against 1, 0, 3, and 2 for the clean model; the five augmentation-only successes lie between 76% and 94% coverage.

V. Discussion

The central observation is a trade-off between performance on clean simulations and the available experimental scans. Augmentation reduced synthetic accuracy while improving measured-pattern top-5 accuracy in every completed fold-0 run pair. Within the primary comparison, the architecture, input size, split, and validation distribution were held fixed. The result is therefore consistent with augmentation reducing sensitivity to some experimental distortions.

The ablations locate that effect inside the augmentation itself. Broadening most strongly affected augmented-validation accuracy, while background had the largest observed effect on experimental counts. This ordering is consistent with the largest differences between our simulations and measurement: simulated peaks use a nominal 0.01° width, and simulated patterns carry no background. The negligible validation changes for small angular shifts and the texture surrogate also depend on the chosen perturbation ranges and implementations. They should not be generalized to large calibration errors or physically resolved preferred orientation. This distinction matters for AlphaDiffract [12], whose authors attribute part of the gap between synthetic and experimental patterns of the same structures to preferred orientation absent from their training perturbations. Our peak-index surrogate produced no measurable validation benefit, which is compatible with that view only if physically resolved texture differs materially from such an approximation.

An initial fold allocation, which assigned space groups from rarest to most common and stopped once the partition quotas were filled, inadvertently left the four most common COD space groups (14, 2, 15, and 19) without training examples, and this explained the shortfall of our early models against the published synthetic benchmark. Adopting the published identifiers resolved the discrepancy, and all results reported here use the corrected split. We describe the pitfall because an allocation that silently omits the head of a strongly skewed label distribution yields results that look plausible but are not comparable to the benchmark.

The class-prior result and modest experimental top-1 accuracy reinforce that this is a candidate-ranking study, not a complete automated structure-identification system. Larger independent evaluations with structure-disjoint splits, broader measured data such as suitable labeled subsets of opXRD [13], and stronger physical perturbation models would test the scope of the observed benefit. Such an evaluation should freeze the method, checkpoints and preprocessing before the new patterns are scored, and report the resulting effect and statistical test as they are; additional independent data can increase statistical power, but significance should be an outcome of that evaluation rather than a target. Those evaluations remain future work. Comparisons with AlphaDiffract [12] or other recent systems would require aligned tasks, labels, data, and metrics before supporting any ranking of methods.

VI. Limitations

Several limits matter for practical use. The 20-pattern collection comes from one source and informed development, and most scans required filling outside their measured range.

Repeated seeds reuse those same observations; they support consistency of direction but do not enlarge the experimental sample. The ablation uses one run per arm, training hardware and precision varied, only one architecture was studied under the corrected protocol, and the augmentation distribution contained one fixed rendering per structure per run. Moreover, structure duplicates inherited from the published split limit what synthetic test accuracy establishes about unseen structures. No full-versus-component experimental comparison reached a nominal 0.05 significance threshold.

VII. Conclusion

A corrected SIMPOD-based Swin-T baseline achieved 82.92% synthetic top-5 accuracy, close to the published reference, enabling a controlled study of augmentation. Full augmentation reduced this score to 79.63% but improved experimental top-5 accuracy from 6/20 to 11/20 in the primary comparison, with smaller positive gains in two additional fold-0 runs, against a class-prior baseline of 7/20. Component ablations point to broadening and background as the most useful interventions under the tested settings, and the gains were not confined to the most extrapolated scans. The evidence supports augmentation for this specific classifier and sample while leaving broad experimental generalization unresolved; testing it requires independent measured patterns scored once on a frozen method.

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